

Using End Behavior and Multiplicity to Sketch Polynomials

Finding zeros and their multiplicity, reading end behavior from the leading term, and building a sketch from both

Directions

- A zero with an *odd* multiplicity is a crossing; a zero with an *even* multiplicity is a turning point on the axis.
- End behavior is decided by two things only: the degree (odd or even) and the sign of the leading coefficient.
- On the sketching problems, plot the zeros and the y -intercept first, then join them so both ends match the end behavior. A smooth freehand curve is fine — exact heights between zeros are not being graded.

1. Factor $p(x) = x^3 - 4x$ completely.

Answer: _____

2. For $p(x) = -2x^3 + 5x - 1$, state $\lim_{x \rightarrow \infty} p(x)$.

Answer: _____

3. Let $p(x) = (x - 1)^2(x + 3)$.

(a) List all real zeros of p .

Answer: _____

(b) Say which zero the graph crosses and which it only touches, and give the reason from the exponents.

Answer: _____

4. Let $p(x) = 3x^4 - x^2 + 2$.

(a) State $\lim_{x \rightarrow -\infty} p(x)$.

Answer: _____

(b) State $\lim_{x \rightarrow \infty} p(x)$.

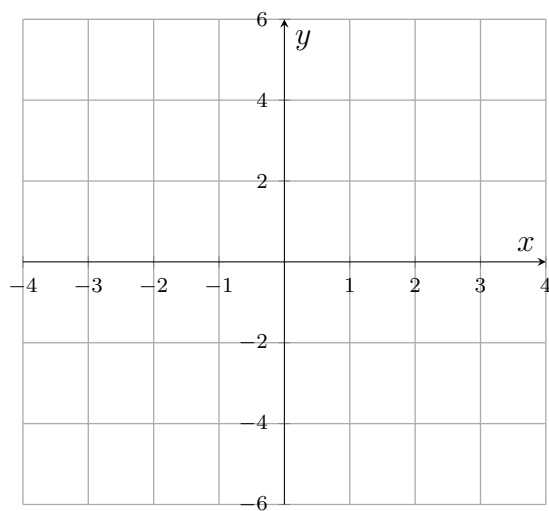
Answer: _____

5. Let $p(x) = (x + 2)(x - 1)^2$.

(a) Find the y -intercept $p(0)$.

Answer: _____

(b) Sketch $y = p(x)$ on the grid. Mark the zeros and the y -intercept.



6. Factor $p(x) = x^4 - 5x^2 + 4$ completely.

Answer: _____

7. Let $p(x) = -x^5 + 3x^2$.

(a) State $\lim_{x \rightarrow -\infty} p(x)$.

Answer: _____

(b) Describe the end behavior of p in words, and name the two features of the rule that decide it.

Answer: _____

8. Let $p(x) = x^3 + 2x^2 - 9x - 18$.

(a) Factor p completely by grouping.

Answer: _____

(b) List all real zeros of p .

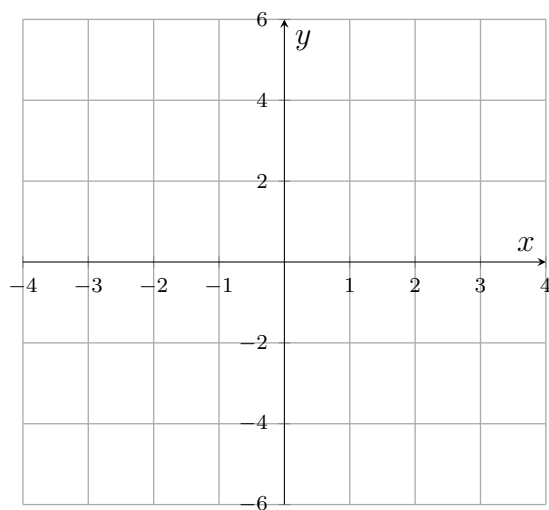
Answer: _____

9. Let $p(x) = x^4 - 4x^2$.

(a) Factor p , then list all real zeros.

Answer: _____

(b) Sketch $y = p(x)$ on the grid, marking each zero as a crossing or a touch.



10. A polynomial p has degree 3, a zero at $x = -1$ of multiplicity 2, a zero at $x = 3$ of multiplicity 1, and $p(0) = -3$. Write $p(x) = a(x + 1)^2(x - 3)$.

(a) Use $p(0) = -3$ to find a .

Answer: _____

(b) Sketch $y = p(x)$ on the grid and state its end behavior at both ends.

